

# Design Report

## Project Overview and Problem Statement

The campus companion app is an application that is designed to help students in their daily college life. Features of the app include:

1. A map to help students find their way around campus.
2. A timetable so students are able to keep up with their daily classes.
3. An events page so students can keep up with new events that they may be interested in.
4. A society events page so students can see events that are happening in any societies that they may be a part of.
5. A lost and found + helpdesk page where students can submit support messages and any lost and found items.

The problem statement that is being addressed is the lack of an app that exists that has all the important information and tools that are needed for students to have an easier college life in one singular place. And the goal of the companion app is for this to be fixed with the previous features that have been mentioned.

## Target Users and User Needs

The targeted users for the campus companion app are first year students and international students. The reason why these are the targeted users is because for first year students the aspect of college can be quite daunting and having a campus companion app can help them settle in easier. The same goes for the international students, they need an easy and simple way to find their way around campus and send support messages if needed, and the campus companion does this perfectly.

A more in-depth example of this would be:

(Persona one) A first-year student is new to the college and struggles to find their way to most of their classes. The solution to this? The map feature on the campus companion app.

Another example:

(Persona 2) The international student struggles to understand one of their lecturers and isn't sure what to do. The solution to this is in the lost and found + helpdesk section of the campus companion app.

## System Design: Inputs, Processes, Outputs

### INPUTS:

-Users Login Details –Users Location Data –Users Timetable Information –Campus Event Details –Society Event Details –Lost and Found information –Help Desk Submission – Accessibility Settings.

### PROCESSES:

-Validating Login Details –Storing User Location Data –Creating Users Timetable – Event Notification Queue –Society Event Notifications Queue –Lost and Found Information Stored –Help Desk Submission stored –Accessibility Settings Saved.

### OUTPUTS:

-Map shows users location on campus –Timetable is personalized for user –Campus Event Notification is sent –Society Event Notification is sent –Lost and Found Information is posted –Help Desk Submission is sent.

This is the system design that allows inputs to process, then become an output.

## Architecture Overview

The front end of the system is a neat and simple layout, made for ease of access for users. This is what the users will be seeing.

All of the data that is stored from the users will be stored in the cloud. This is what allows all users to be able to save their personalized choices and settings on the campus companion app.

The back end of the system is not seen by the users at all. It contains the code that allows for the users to access their timetables and features like the map. It basically is the part of the system that processes the requests from the users.

## Data Design

The data design section of the campus companion app was made to improve the accuracy of the event recommendation system.

What happens is it takes a student as a sample and shows the events that they are interested in; after doing this it shows the events that they are interested in it then uses this information to grab other users that are also interested in the same events, its will then recommend these events that they are interested in to the sample user.

After this, there is a predicted click probability feature. This feature shows the chance of the sample user actually clicking on these recommended events.

The system, however, will not collect and save personal details of the users. This includes home addresses, finances, passport details, and social security numbers. The app only uses details provided by the user that are necessary for example temporarily using the users location for the map feature.

## Usability and UI design Decisions

The interface for the campus companion app is simplistic and easy to use. It's made this way, so users are able to use it fast and get what they need to do done. The main dashboard is comprised of the most important features of the campus companion app being: The timetable feature, the map feature, the events feature, the society events feature. And the lost and found + helpdesk feature. There is also a setting and account management feature.

The UI design keeps a simple and clean theme for each page of the companion app so that it has a clean appearance and so there are no overwhelming aspects of the app.

The buttons to access different features of the app are distinct from each other for ease of uses, there is also a description under each heading for example the timetable description explains that the timetable is a “Weekly academic timetable with group selection and week navigation” and the events description “Browse campus events, view details, and save your interest”. This is useful for users that might not speak English well as its written in clear language.

The notification system on the campus companion app is also very useful for users as for example when a user joins a society the events from the society are automatically sent out as notifications where the user doesn't have to do anything to get them sent to them.

## Accessibility Plan

The accessibility plan was created based on the WCAG checklist (Web Content Accessibility Guide). This allowed us to review the accessibility of our website, web content, and web applications. We used the four main principals of WCAG to do this, being: Perceivable, Operable, Understandable, and Robust.

For the perceivable principal, we made sure that the campus companion app has adjustable font sizes so that users that may be visually impaired can also use the app to the fullest, as well as a button that reads out loud for the user and a contrast button to help visibility.

For the operable principal, we made sure that the campus companion app would be usable through touch screen, mouse, and keyboard. Of which the keyboard commands are located in the settings tab, which explains what the different keyboard commands do, for example “Use Enter or Space to activate buttons or links”.

For the understandable principal, we made sure to add descriptions underneath heading such as Map and Timetable explaining what they do for ease of understanding for the users. If the app doesn’t contain an easy explanation on how to use the feature, there is also the helpdesk feature which allows the user to submit support questions so they can sort out any issues they are having.

Finally, for the robust feature we made sure that the campus companion app would be built for long term use as it is made a large demographic and is reliable. The code behind it is strong and it is compatible with new technologies that are coming out such as assistive tools.

## Privacy, Security, and Compliance

The campus companion app follows the GDPR(General Data Protection Regulation) basics, being the lawful basis conceptually, minimization, retention; risks+ mitigations.

The campus companion app does not use any data collected from the user without consent, and any data in the first place that is collected is only that of necessary data that is related to the optimal use of the campus companion app.

All data that is used is encrypted, which is a good security measure to ensure that no user data is leaked.

## Deployment Plan

Before the campus companion app is officially deployed, it needs to go through some tests. Its base version will be tested by volunteers who test to see if there are any issues with the app. Once this process is complete and any issues found are fixed, the app will be released onto Netlify so users can use it.

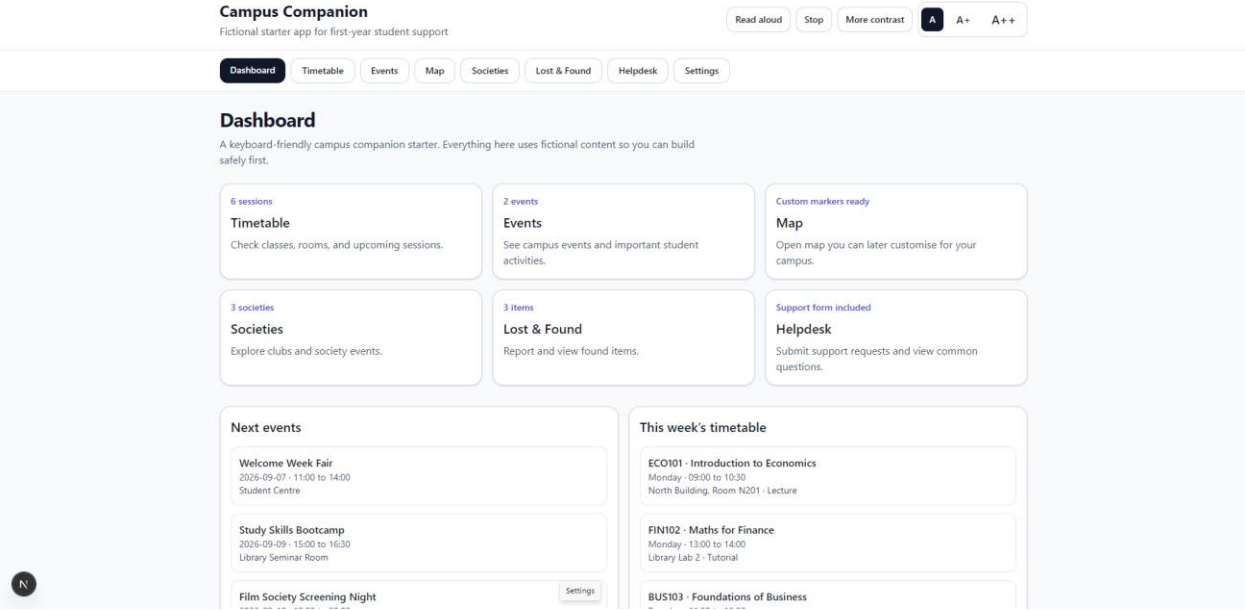
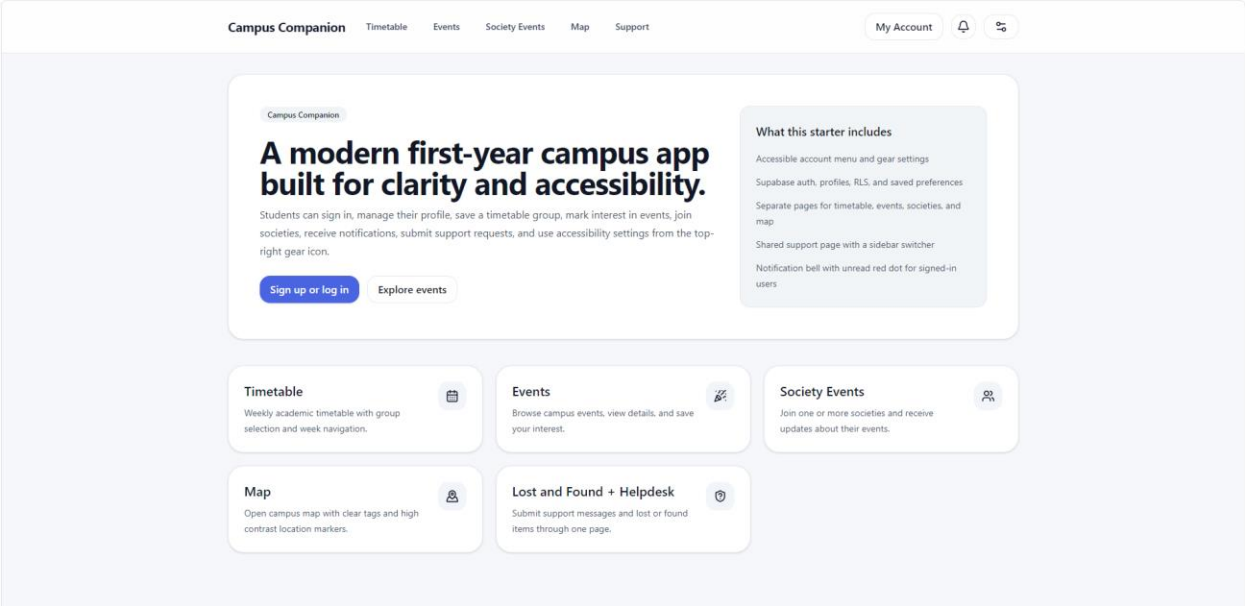
After the campus companion app is deployed, the development of it isn't finished; it will need to be updated constantly to stay smooth; it will also need to add more features that

may have been overlooked in development. There will also have to be a maintenance team to make sure it stays up and healthy at all times.

## Appendix

The following images are examples from the Usability and UI Design decisions.





Campus Companion

Fictional starter app for first-year student support

Read aloud

Stop

More contrast

A

A+

A++

Dashboard

Timetable

Events

Map

Societies

Lost & Found

Helpdesk

Settings

### Timetable

Simple weekly timetable view. Replace the fictional modules with your own Supabase data later.

Monday

ECO101 - Introduction to Economics

North Building, Room N201

09:00 to 10:30 - Lecture

FIN102 - Maths for Finance

Library Lab 2

13:00 to 14:00 - Tutorial

Tuesday

BUS103 - Foundations of Business

South Building, Room S108

11:00 to 12:30 - Lecture

Wednesday

DAT104 - Digital Skills for Study

Innovation Hub

10:00 to 12:00 - Lab

Thursday

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### Campus events

General campus events for first-year students.

2026-09-07

Welcome Week Fair

11:00 to 14:00 - Student Centre

Meet campus teams, support services, and student leaders.

2026-09-09

Study Skills Bootcamp

15:00 to 16:30 - Library Seminar Room

Practical tips on note taking, planning, and revision.

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### Campus map

OpenStreetMap based starter map. Later, replace the markers and coordinates with your own campus data.



Map locations list

Keep this list even when you customise the map later. It gives keyboard and screen reader users a reliable alternative.

Student Centre

Support

Main student services, social space, and event desk.

Coordinates: 53.3496, -6.2602

Campus Companion

Fictional starter app for first-year student support

Read aloud Stop More contrast A A+ A++

Dashboard Timetable Events Map Societies Lost & Found Helpdesk Settings

Societies

List of fictional societies and their events.

Society directory

**Coding Society**

Build projects, learn web development, and meet other beginners.

Meets Wednesday at 17:00  
Innovation Hub  
coding.society@fictional-campus.test

**Debate Society**

Friendly debates for confidence, public speaking, and fun.

Meets Tuesday at 16:00  
Student Centre, Room 4  
debate.society@fictional-campus.test

**Film Society**

Screenings, reviews, and casual discussion nights.

Meets Thursday at 18:00  
Media Room  
film.society@fictional-campus.test

Society events

Film Society

**Film Society Screening Night**

2026-09-10 - 18:00 to 20:00  
Media Room

A relaxed first screening with discussion afterwards.

Coding Society

**Coding Society Build Session**

2026-09-12 - 17:00 to 19:00  
Innovation Hub

Beginner friendly project session for web app ideas.

N

Campus Companion

Fictional starter app for first-year student support

Read aloud Stop More contrast A A+ A++

Dashboard Timetable Events Map Societies Lost & Found Helpdesk Settings

Lost and found

View found items and report new ones using fictional details only.

**Current items**

**Iphone 15**

Tech - Main Building, Room 224

Iphone 15 with a red phone case

Status: open  
Alias: Mayank

**Blue Water Bottle**

Personal Item - Library, Floor 1

Metal bottle with sticker on the side.

Status: open  
Alias: Campus Desk

**Grey Hoodie**

Clothing - North Building, Room N201

Medium size hoodie found after lecture.

Status: open  
Alias: Campus Desk

**Report an item**

Use an alias instead of real personal details.

**Item name**

**Category**

Select a category

**Found location**

**Description**

**Contact alias**

Example: Green Backpack Owner

**Submit report**

N

Campus Companion

Fictional starter app for first-year student support

Read aloudStopMore contrastA A+ A++

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## Helpdesk

Support form and common questions for first-year students.

### Frequently asked questions

► Where do I get help with my timetable?

► How do I report a lost item?

► Can I change text size and contrast?

### Contact helpdesk

Keep all requests fictional while you build and test.

Request title

Category

Select a category

Details

Send request

N

Campus Companion

Fictional starter app for first-year student support

Read aloudStopMore contrastA A+ A++

Dashboard

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Lost & Found

Helpdesk

Settings

## Accessibility settings

Use these controls to make the interface easier to read and navigate.

### Controls

These settings stay saved in local storage for this browser.

Read aloud

Stop

More contrast

A A+ A++

### Keyboard tips

Use Tab and Shift + Tab to move through controls.

Use Enter or Space to activate buttons and links.

Use the skip link at the top to jump to main content.

Keep the text list under the map for non pointer users.

N

The following image is in relation to the Data Design section.

```
Loaded event interaction data rows: 3026
Unique users: 500
Unique events: 10
Columns available: ['user_id', 'event_id', 'name', 'tag', 'interested', 'category', 'created_at']

MODEL SUMMARY
Features used for user similarity:
- Interest counts by category (academic, social, recreational, wellbeing)
- Total events interested in
- Diversity of interests (number of different categories)
- Interest ratio (interested events / total shown events)
Model choice: An event recommender system using K-Nearest Neighbors and Logistic Regression. KNN finds users with similar interests
ve filtering, while Logistic Regression predicts click likelihood based on user and event features.

Sample user: STU0001
Total events shown: 6
Events interested in: 4
Interest categories: ['academic' 'recreational']

Similar users:
1. User STU0242 (similarity: 1.00)
  Events interested: 4, Categories: ['academic' 'recreational']
2. User STU0096 (similarity: 1.00)
  Events interested: 4, Categories: ['academic' 'recreational']
3. User STU0150 (similarity: 1.00)
  Events interested: 4, Categories: ['academic' 'recreational']

Collaborative Filtering Recommendations:
1. Debate Competition (Academic) - academic
2. Sports Tournament (Sports) - recreational
3. Art Exhibition (Arts) - recreational

Hybrid Recommendations (KNN + Logistic Regression):
1. Debate Competition (Academic) - academic
  Predicted click probability: 0.76
2. Sports Tournament (Sports) - recreational
  Predicted click probability: 0.55
3. Art Exhibition (Arts) - recreational
  Predicted click probability: 0.55
```

## The WCAG checklist

|                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                               |                                                                                                                                                                                         |                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                                                                                                              |                                                                                                                                            |                                                                                                      |                                                                                                                                                                 |
| <b>Perceivable</b>                                                                                                                                                                                                                                                              | <b>Operable</b>                                                                                                                                                                                                               | <b>Understandable</b>                                                                                                                                                                   | <b>Robust</b>                                                                                                                                                                                                                                        |
| Information and user interface components must be presentable to users in ways they can perceive. For example, it's important to present information that can be perceived in different ways, where a user can adjust color contrast or font size, or view captions for videos. | User interface components and navigation must be functional for users in ways they can operate. For example, a user must be able to perform required interactions using a keyboard or voice commands, not just using a mouse. | Information and user interface operation must be understandable. For example, information and instructions should be clear and navigation methods should be easy to understand and use. | Content must be robust enough that it can interpreted reliably by a wide variety of users and assistive technologies. As technologies evolve, code and content should remain accessible for users of common and current assistive devices and tools. |

|                   |                |
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| Student number    | A00027470      |
| Group name/number | 22             |

|             |                                                                                                                                                                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Module      | Nature of Enterprise Computing                                                                                                                                                                                                                         |
| Assessment  | CA3 Design Report (Component 1)                                                                                                                                                                                                                        |
| Declaration | I confirm that I did not/will not use generative AI (e.g., ChatGPT/Claude or other genAI tools) to produce any part of the design report. I understand that genAI use in this component results in a mark of 0 for this component for the whole group. |
| Signature   | Jack McGurrian                                                                                                                                                                                                                                         |
| Date        | 26/04/2026                                                                                                                                                                                                                                             |

|                   |                                                                                                                                                                                                                                                        |
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| Group name/number | 22                                                                                                                                                                                                                                                     |
| Module            | Nature of Enterprise Computing                                                                                                                                                                                                                         |
| Assessment        | CA3 Design Report (Component 1)                                                                                                                                                                                                                        |
| Declaration       | I confirm that I did not/will not use generative AI (e.g., ChatGPT/Claude or other genAI tools) to produce any part of the design report. I understand that genAI use in this component results in a mark of 0 for this component for the whole group. |
| Signature         | Dean Nolan                                                                                                                                                                                                                                             |
| Date              | 26/04/26                                                                                                                                                                                                                                               |

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| Group name/number | 22                                                                                                                                                                                                                                                     |
| Module            | Nature of Enterprise Computing                                                                                                                                                                                                                         |
| Assessment        | CA3 Design Report (Component 1)                                                                                                                                                                                                                        |
| Declaration       | I confirm that I did not/will not use generative AI (e.g., ChatGPT/Claude or other genAI tools) to produce any part of the design report. I understand that genAI use in this component results in a mark of 0 for this component for the whole group. |
| Signature         | Mayank Singh                                                                                                                                                                                                                                           |
| Date              | 26/04/26                                                                                                                                                                                                                                               |